

Replacing silicon with shape shifting molecules

Researchers at the Indian Institute of Science's Centre for Nano Science and Engineering (CeNSE) have demonstrated molecular electronics that could move computing beyond

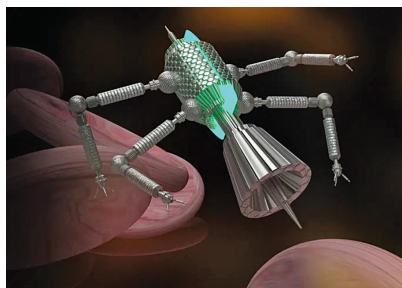


Research team (Credit: CeNSE, IISc)

silicon. The team developed shape-shifting molecular devices that can dynamically function as memory, logic, or synaptic elements by tuning their chemistry. By embedding computation directly into material structure, the approach merges storage and processing at the molecular scale, opening a path to ultra-compact, low-power hardware for future computing and AI systems.

Seeing medical microrobots in real time

A new MRI-based imaging approach is enabling real-time control of magnetic medical microrobots, addressing long-standing delays and image artifacts that limited live



3D rendering of a microbot (Credit: <https://interestingengineering.com>)

navigation during procedures. The technique, developed by scientists at Huazhong University of Science and Technology, redesigns MRI sequences using multi-frequency dual-echo scanning to cut repetition

times to around 30ms. This allows near-real-time tracking with less than 1% error while using MRI for both imaging and actuation, advancing prospects for precise, minimally invasive therapies such as targeted drug delivery.

All-optical AI chip challenges GPUs

A new all-optical AI processor points to a potential alternative to conventional silicon GPUs, using light rather than electrons to run generative AI models at extreme speed

and energy efficiency. Dubbed LightGen, the system was developed by research teams from Shanghai Jiao Tong University and Tsinghua University. Prototype demonstrations showed up to 100x faster processing and 100x lower energy consumption on image and video generation tasks compared with GPUs such as Nvidia's A100, enabled by more than two million photonic 'neurons' and 3D-stacked optical architectures. While still at the research stage, scaling LightGen for large, general-purpose AI workloads remains a key challenge.

AI shields wireless networks

An AI-driven security system has been developed to protect wireless networks against real-time jamming attacks, improving the reliability of mission-critical communications. The platform uses a dual-agent AI framework that continuously learns interference patterns and autonomously



UOttawa researchers discussing the anti-jamming application (L to R: Ghazal Asemian, Dr Iman Amini, Prof. Burak Kantarci) (Credit: Professor Burak Kantarci, University of Ottawa)

adapts control decisions to sustain connectivity under active radio disruption. Developed by a research team at the University of Ottawa, the system acts like a 'digital immune response' for wireless infrastructure used in emergency response, healthcare, and smart cities, with early tests showing strong performance in hostile radio conditions.

Reinforcement learning speeds optical AI learning

Reinforcement learning has been shown to significantly accelerate training in optical AI hardware by enabling model-free, on-device optimisation. Researchers at UCLA applied the Proximal Policy Optimisation (PPO) algorithm directly to physical optical processors, avoiding errors common in simulation-based training. The method successfully performed tasks such as focusing light through diffusers, generating holograms, correcting optical aberrations, and classifying handwritten digits. By reducing the number of experimental runs, this approach makes optical AI systems more efficient, adaptive, and robust for practical, real-world applications.